

CAPSTONE PROJECT PROPOSALS

Presented by Malak Adel

TOPIC 1: CREDIT CARD FRAUD DETECTION

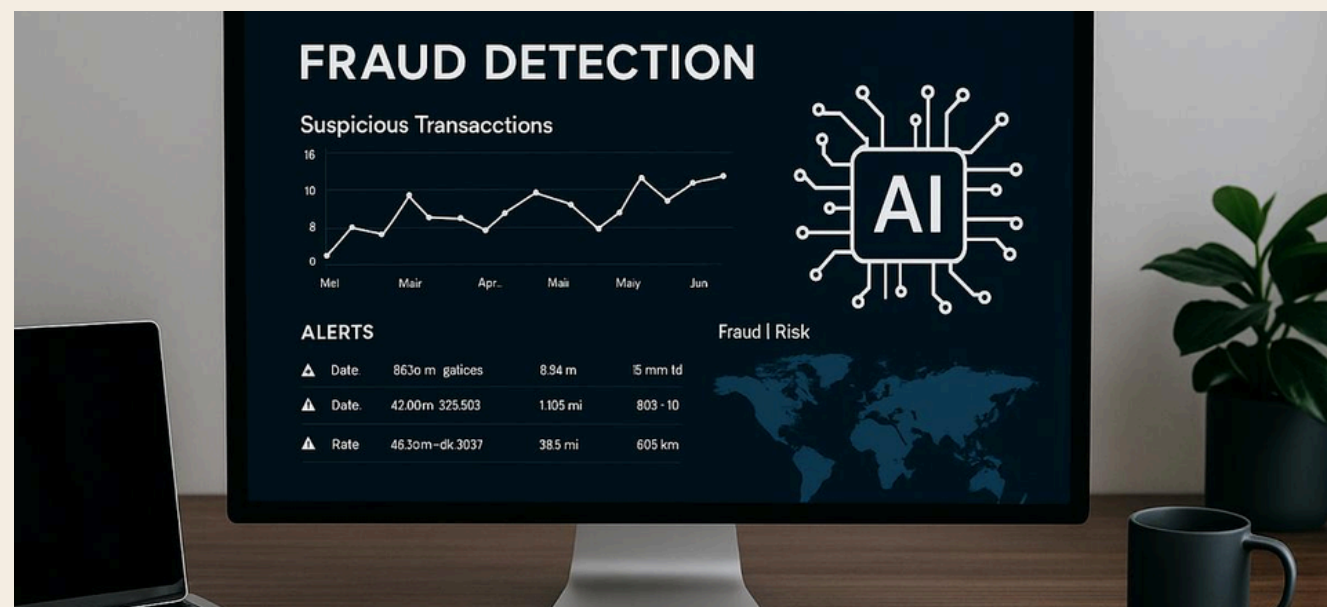
GOAL & PROBLEM

- Financial institutions lose billions due to fraudulent transactions.
- Build a classification model to predict `is_fraud`.
- Detect fraud accurately while reducing false alarms.



EDA FINDINGS

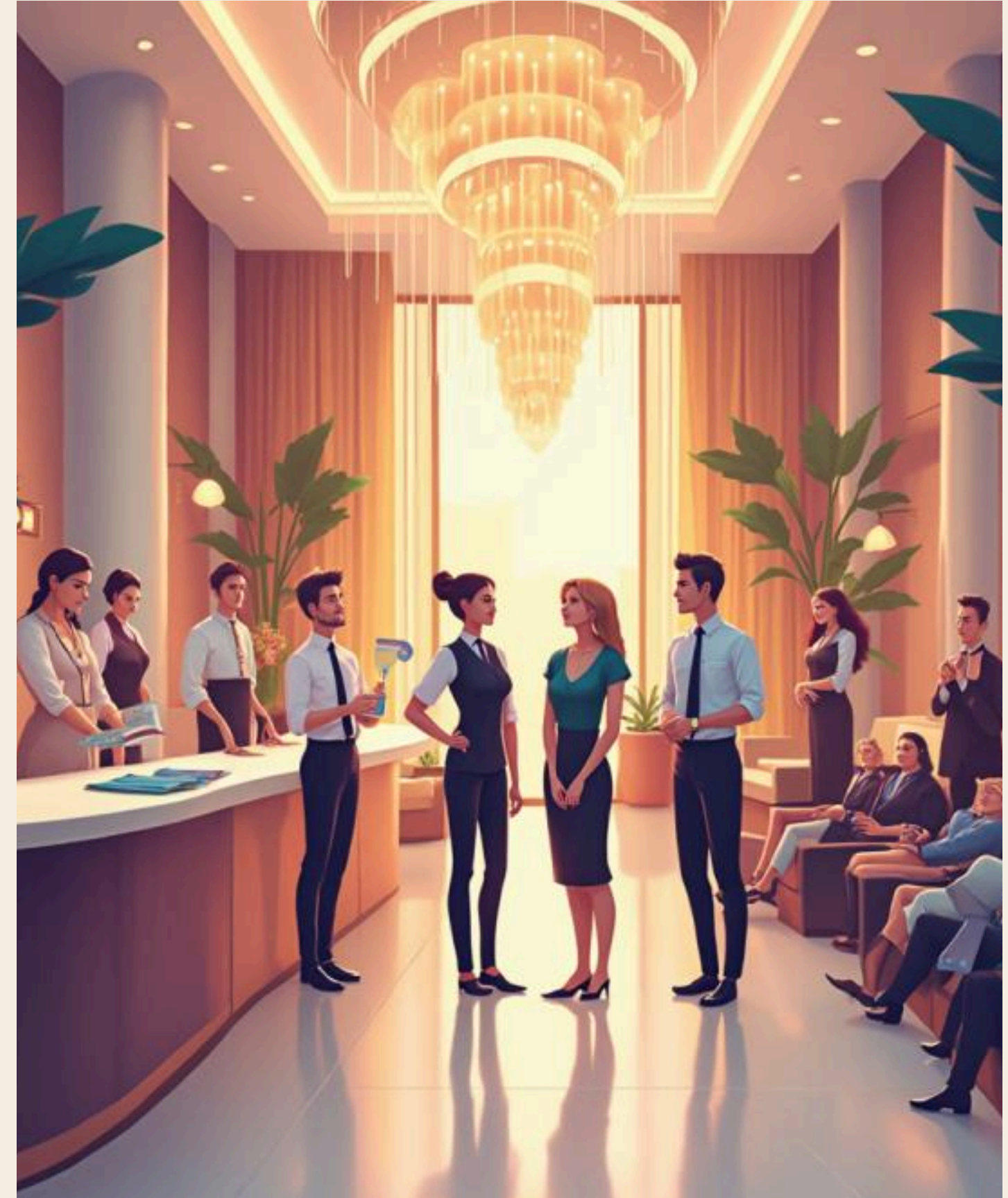
- Target: is_fraud
- Problem Type: Classification
- Missing Values: None
- Numerical Columns: 6
- Categorical Columns: 9
- Outlier: amt (Transaction Amount)



TOPIC 2: HOTEL BOOKING DEMAND

GOAL & PROBLEM

- Predict hotel booking cancellations.
- Support occupancy planning and dynamic pricing.
- Reduce revenue loss from cancellations.



EDA FINDINGS

- Target: IsCanceled
- Problem Type: Classification
- Missing Values: Country (488), Children (4)
- Numerical Columns: 19
- Categorical Columns: 11
- Outliers: ADR, LeadTime

TOPIC 3: REAL ESTATE PRICE PREDICTION

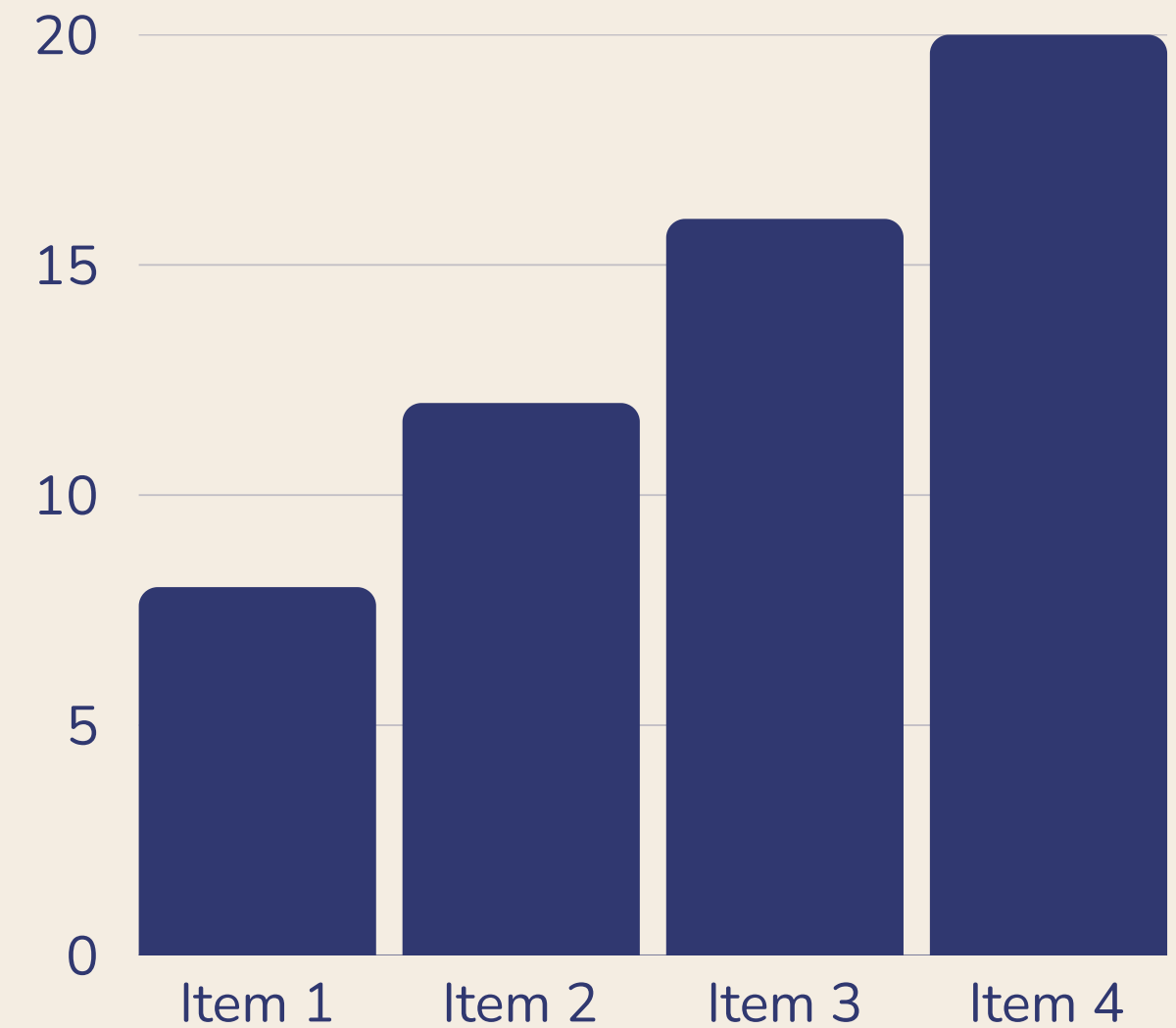
GOAL & PROBLEM

- Predict house sale prices using historical data.
- Improve valuation accuracy.
- Support buyers, sellers, and agents.



EDA FINDINGS

- Target: price
- Problem Type: Regression
- Missing Values: None
- Numerical Columns: 18
- Categorical Columns: 2
- Outliers: price, sqft_lot, bedrooms



THANK YOU